

ARTICLE

Coastal and Marine Ecology

Harnessing trophic cascades to improve foundation species restoration: A meta-analysis

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Abstract

Global declines in foundation species and associated losses of biodiversity and ecosystem services have made restoration of foundation species and the habitats they create a top priority. Because restoration success has been highly variable, the field is increasingly using both experimentation and ecological theory to augment success. We conducted a meta-analysis of studies (18) in estuarine and marine systems, which increased the abundance of top predators as a restoration approach, thus improving biogenic habitat-forming foundation species recovery by reducing consumers. Beyond testing the general idea of whether top predator addition enhances foundation species restoration, we examined whether the degree of restoration success could be explained by classic predictors of top-down strength, such as taxonomic identity of the foundation species and top predators, direct and indirect effects of top predators, and the latitude and duration of the study. Top predators successfully elicited a trophic cascade by decreasing consumer abundances (direct effect) and increasing foundation species recovery (indirect effect) when compared with control restoration projects without top predators. Direct effects are expected to be stronger than indirect effects, in accordance with classic ecological theory. We found no evidence of magnitude differences between direct and indirect effects of top predator addition. Increasing invertebrate top predators produced a 3.5× greater increase in the basal trophic level than adding vertebrate top predators. Despite the strong effect of predator type, the log response ratio of all foundation species types (algae, animal, and plant) increased 10-fold in top predator addition treatments. We found diminishing effects of top predator addition as study duration increased, which supports recent developments that the positive effects of marine protected area designation decline over time. Our results highlight the promise of harnessing top-down control via top predator addition as a mechanism to aid foundation species recovery.

KEYWORDS

foundation species, meta-analysis, restoration ecology, top-down control

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INTRODUCTION

In coastal and marine systems, foundation species such as bull kelp algae, oysters, mangroves, and seagrasses form habitats that promote local biodiversity by providing refugia, serving as a food source, and ameliorating environmental (e.g., temperature) and biological stressors (e.g., predation; Angelini et al., 2011; Duarte et al., 2013; Lamy et al., 2020; Smith et al., 2022). Globally, many of the ecosystem functions provided by these foundation species are valued as important ecosystem services, including blue carbon storage, nutrient cycling, and coastal protection from erosion (Dobson et al., 2006). However, due to habitat depletion, overharvest, and anthropogenic climate change, these foundation species have declined with cascading negative impacts on biodiversity, ecosystem functioning, and fisheries yields (Bromberg Gedan et al., 2009; Hoegh-Guldberg, 1999; Newell, 1988). Consequently, the restoration of foundation species is now a global top priority for both scientists and government agencies. Worldwide, the United Nations has embarked on a “Decade of Ecosystem Restoration” with 50 founding initiatives taking place from 2021 to 2030, four of which focus on oceans and coasts, including one project that aims to restore 2800 ha of rice fields to mangrove habitat in Guinea-Bissau (Dickson et al., 2021). In the United States, as part of the 2009 American Reinvestment and Recovery Act, the federal government allocated US \$154.1 million to coastal and marine restoration projects (Samonte et al., 2017).

Despite these significant investments, consistent success in restoring coastal and marine foundation species remains elusive. For example, in a recent review of the native Pacific coast oyster (*Ostrea lurida*), restoration ecologists found that nearly half of the restoration projects were associated with declines in adult oyster density five years post-restoration (Ridlon et al., 2021). Similar results were associated with temperate seagrass restoration where the median survival of replanted seedlings was only 38% despite a median restoration cost of over US \$100,000 per hectare (Bayraktarov et al., 2015). In these same systems, a common element in reduced success restoration projects is run-away consumption (Bertness et al., 2014; Silliman & Bertness, 2002). In San Francisco Bay, when *O. lurida* restoration outplants were caged to exclude the carnivorous whelk (*Urosalpinx cinerea*), oyster survival increased from 0% to 1.6% to 89% (Cheng et al., 2022). Meanwhile, in Chesapeake Bay, restoration outplants of the submerged aquatic vegetation (SAV) wild celery (*Vallisneria americana*) that excluded herbivores increased SAV survival by 50% (Moore et al., 2010).

In these examples, consumer pressure could be limiting success for two reasons. First, ambient environmental conditions may favor the consumer over the foundation species

by weakening defense mechanisms and increasing susceptibility to consumption. Examples include studies of ocean acidification showing decreased shell strength in oysters, although their consumers (shell-forming carnivorous whelks) were unaffected by increased $p\text{CO}_2$ (Sanford et al., 2014). Similarly, drought conditions were found to decrease plant health and increase grazing susceptibility in salt marshes (Silliman et al., 2005). A second limitation of restoration success may concern the loss of marine predators that historically benefitted foundation species by controlling populations of the consumers (Dobson et al., 2006; Vizzini et al., 2017). In the classic sea otter-sea urchin-kelp example, the loss of sea otters released the intermediate sea urchin consumer, which resulted in major kelp losses (Estes & Palmisano, 1974). Thus, improving restoration success may hinge on addressing one or both of these consumer effects, rather than solely focusing on outplanting the focal foundation species, and we propose that successful foundation species restoration may also depend upon the field adopting a comprehensive view of the problem, moving beyond an exclusive focus on the foundation species themselves to a consideration of the ecosystem as a whole.

While local-scale restoration efforts typically cannot alter ambient environmental conditions, we are aware of restoration efforts that manipulated the presence of top-predators in an attempt to dampen consumer pressure and thus increase foundation species recovery, where they compared juvenile *O. lurida* survival in the presence and absence of large adult crabs, *Cancer* spp. (Grason & Buhle, 2016). Detecting indirect positive benefits of predators on basal resources (trophic cascades [TCs]) has long been studied in ecology, but testing whether they can be consistently harnessed to promote foundation species recovery is less certain. Restoration managers have long turned to ecological concepts to increase success, and theory general to ecology and specific to TCs abound for predicting success of restoration TCs (Fodrie et al., 2014; Wainwright et al., 2017). For example, the effects of top predators have been shown to typically attenuate with trophic level, that is, the direct effects are typically stronger than the indirect effects. Are restoration experiments characterized by this trophic attenuation or do they generally succeed by having strong direct and indirect effects? (Power, 1992; Schmitz et al., 2000; Shurin et al., 2002). Identity may also affect the strength of the TC; foundation species such as low-carbon, short generation time autotrophs (plants and macroalgae) may respond more strongly to TC experiments than heterotrophic corals and oysters (Trussell et al., 2017). Additionally, previous meta-analyses on TCs suggest that vertebrate top predators induce stronger responses in foundation species than invertebrate predators (Shurin et al., 2002; Steneck et al., 2004). Another trait to consider when interpreting variability in successfully harnessing TCs in

restoration efforts concern predictions from general ecology about location, where predation pressure typically increases with decreasing latitude (Freestone et al., 2013; Pennings et al., 2009). In conclusion, ecological drivers such as interaction strength, latitude, and identity of foundation species and top predators may influence the success of any given foundation species restoration project.

In addition to these ecological drivers, several methodological factors may influence top predator addition as a restoration technique, including manipulation type, predator density, and study duration. In order to expand the sample size, mensurative, natural top predator addition through harvest cessation or MPA designation, were included in the meta-analysis. Previous work has predicted that smaller, more controlled experiments result in stronger responses than larger, natural experiments (Englund & Cooper, 2003; Sagarin & Pauchard, 2009). Therefore, we might expect that pressed restoration projects have a greater foundation species response than mensurative projects. Additionally, we considered whether predator density would affect foundation species response because more predators could reduce the consumer population more quickly or thoroughly. Finally, a recent development in MPA theory states that as time since MPA designation increases, the positive benefits on foundation species decline (Magdaong et al., 2014). Methodological drivers may also contribute to variation in top predator addition as a foundation species restoration technique, thus we included them in our analysis.

To examine the success of these individual restoration projects and of top predator addition as a whole, both ecological and methodological drivers need to be analyzed in order to make tractable recommendations to restoration managers. We conducted a meta-analysis of estuarine and marine foundation species restoration projects where we tested the following hypotheses: (1) top predators decrease consumers and increase foundation species; (2) the direct effect of top predator on consumer will be stronger than the indirect effect on foundation species; and (3) foundation species response are influenced by the context of the restoration attempt, such as the location, duration, manipulation type, and taxonomic identity of the top predator and foundation species. Due to the papers available, the foundation species included are generally more habitat formers than food sources.

METHODS

Data sources

To evaluate whether top predators influence the restoration success of foundation species, we conducted a

meta-analysis of peer-reviewed experiments published between December 1985 and January 2021. Our study included several searches focused on titles, keywords, and abstracts in the *Web of Science* database using the following search terms for (1) habitat type (coastal, estuary, intertidal, marine, salt marsh, sea, and ocean), (2) trophic level (trophic cascade, top predator, and apex predator), and (3) restoration (restor* and constr*). Of the resulting papers that met our search criteria, we examined subsequent papers that cited them to find additional potentially relevant publications. To locate papers potentially missed from our search, we also evaluated the bibliographies of other reviews and meta-analyses that addressed broader trends in restoration ecology and top-down control (Heck & Valentine, 2007; Micheli et al., 2004; Pinnegar et al., 2000; Ritchie & Johnson, 2009; Shaver & Silliman, 2017).

To be included in this meta-analysis, a publication had to satisfy three criteria: (1) the restoration effort occurred in a marine or estuarine system; (2) the restoration effort focused on autotrophic (kelp, salt marsh, and seagrass) or heterotrophic (coral and oyster) foundation species that can form critical biogenic habitat; (3) the restoration effort manipulated a top predator and observed the responses of the foundation species as well as intermediate consumers of the foundation species (hereafter, consumer). Publications must also have included metrics that allowed quantifying changes in at least three trophic levels (top-predator, intermediate consumer, and foundation species).

This meta-analysis focused on marine and estuarine restoration publications (criterion one) because previous reviews revealed that top-down control in these systems is not only generally strong, but also highly variable (Cebrian, 1999; Shurin et al., 2002, 2006), thereby providing a large range of possible outcomes of top-down control on restoration success. Meanwhile, criterion two was necessary in order to exclude studies focused primarily on quantifying the influence of top-down control, regardless of whether or not they occurred with an explicit goal of restoring a foundation species (i.e., via reestablishment or introductions of top predators). Finally, criterion three was necessary to exclude studies that may have simulated the consumptive effects of top predators by simply excluding or removing the consumer of the foundation species. This criterion enabled the quantification of both the direct and indirect effects of top predators in the restoration studies.

Although top predator abundance itself had to be manipulated within the experiment and compared with a corresponding control restoration project without top predators, the manipulation of top predator abundance could occur in two ways. The researchers could use a

pressed experiment, in which case top predators were increased or excluded from the focal site. Alternatively, researchers could use a mensurative approach where top predator abundance was increased by establishing protected areas designed to decrease the disturbance and harvesting of top predators (e.g., MPAs). For the mensurative approach, either time (i.e., sampling prior to MPA designation) or location (area outside protected area) could be utilized as a control. After reviewing 2168 titles and 183 abstracts, we found 18 publications that explicitly met our criteria and were suitable for inclusion in this meta-analysis (Table 1).

For each publication, we extracted the mean response of both the consumer and foundation species in the control and treatment levels. We employed the open-source software WebPlotDigitizer (version 4.4) to retrieve the mean values from the figures. Due to the variety of foundation species, there was no limitation on which metric the study used to measure foundation species recovery. If the paper presented multiple response metrics, such as including density, percent cover, growth, and survival, we used the percent cover metric because it was the most frequently recorded response across studies. However, not all studies presented multiple metrics; in those cases, whatever response metric they quantified was used. Due to the large scale of the mensurative studies, a lack of repeatability, and variation in response metrics, SE was not included. In mensurative studies where the authors compared multiple MPAs, we used the average of all MPAs; if this was not available, the data from the largest MPA were used. If the paper examined multiple techniques other than top-predator manipulation (e.g., nutrient addition), we used the data from the top-predator only and the control treatments and excluded treatments where additional restoration methods were employed. If the results were reported as a time series, we used the last data point available to ensure consistency with papers that only presented data from one time point. Lastly, we also collected data on factors that frequently influence the context-dependency of top-down control including study duration, species of each trophic level, latitude of the study, and whether the manipulation was pressed or mensurative for each publication.

Data analysis

To promote comparisons across studies, we calculated the log response ratio (LRR) of each trophic level in the top predator addition treatment relative to the control (no/low) top predator abundance treatment (O'Dea et al., 2021). LRR was chosen because it is an easily interpretable metric (the proportional change in focal species across studies) and because it lacks specific assumptions

about variance (in contrast to Hedge's d), thereby allowing for the inclusion of more studies (Shurin et al., 2002). Following the guidelines set out by Hedges et al. (1999), we calculated the LRR for the consumer and foundation species in the top predator addition relative to the control; two LRRs were extracted per paper, one for the intermediate consumer and one for the foundation species (36 total). A positive LRR indicates an increase in the focal species in top predator addition and a negative LRR signifies a decrease in the focal species in the top predator addition treatments. In almost all studies, a positive LRR for the foundation species is considered as predator-mediated restoration success, for example, top predator addition increased foundation species response. However, in some studies due to the metric used, such as coral tissue loss, a negative LRR for the foundation species (reduced loss) is also considered a predator-mediated restoration success (Delgado & Sharp, 2020; Hill & Heck, 2015; Mumby et al., 2006). In these cases, we multiplied the foundation species LRR by -1 .

Statistical analysis

All statistical analyses were conducted in R statistical software (version 4.2.1; R Core Team, 2017). The residuals of the LRR were gamma-distributed, so we used nonparametric analyses to test differences between categorical response variables (direct and indirect effects, foundation species and predator identities, manipulation type, and treatment) in the "stats" package. Direct and indirect effects were calculated by comparing the magnitude (i.e., the absolute value) of the LRR of the consumer (direct) and foundation species (indirect). For discrete response variables (latitude, predator magnitude, and duration), we used generalized linear models (glm2 package) with the main effect being the LRR of the foundation species. Predator magnitude change was quantified by calculating the percent increase in top predator density from control to treatment. While two LRRs were extracted per paper, they were never included in the same analysis (i.e., all foundation species LRR were analyzed separately from the consumer LRR), so no random effects were required because each paper could be considered an independent sample. Post hoc power analyses were conducted in GPower software (version 3.1.9; Faul et al., 2007). Outliers were identified and removed based on the methodology in Faraway (2016) by calculating Cook's distance and leverage.

RESULTS

All of the following results are presented on a log-scale: an LRR of 1 represents a 10-fold increase in the focal

TABLE 1 Summary of publications used in our meta-analysis.

Citation	FS	FS type	TP	TP type	Lat	No. trophic levels	Manip
Barrett et al. (2009)	Macroalgae	Al	<i>Jasus edwardsii</i> , Rock lobster	I	−42.4	3	M
Castilla and Dúran (1985)	<i>Jehlius cirratus</i> , Barnacle	An	<i>Concholepas concholepas</i> , Carnivorous snail	I	−33.3	3	M
Delgado and Sharp (2020)	<i>Acropora cervicornis</i> , Staghorn coral	An	<i>Thais deltoidea</i> , Rock snail	I	24.9	3	Pr
Dúran and Castilla (1989)	Macroalgae	Al	<i>C. concholepas</i> , Carnivorous snail	I	−33.3	3	M
Edgar and Barrett (1999)	<i>Ecklonia radiata</i> , Spiny kelp	Al	<i>Latridopsis forsteri</i> , Bastard trumpeter	V	42.4	3	M
Galasso et al. (2015)	<i>Cystoseira</i> spp., Kelp	Al	<i>Marthasterias glacialis</i> , Seastar + <i>Thalassoma pavo</i> , Ornate wrasse	I + V	38.4	3	M
Grason and Buhle (2016)	<i>Ostrea lurida</i> , Pacific oyster	An	<i>Cancer</i> spp., Rock crab	I	47.2	3	Pr
Giudetti (2006)	<i>Cystoseira</i> spp., Kelp	Al	<i>Coris julis</i> ; Mediterranean wrasse; <i>Diplodus sargus</i> , <i>vulgaris</i> , White, Two-sided seabream	V	40.4	3	M
Hill and Heck (2015)	<i>Halodule wrightii</i> , Shoal grass; <i>Thalassia testudinum</i> , Turtle grass	P	<i>Ardea herodias</i> , Great blue heron	V	30.2	4	Pr
Hughes et al. (2013)	<i>Zostera marina</i> , Eelgrass	P	<i>Enhydra lutris</i> , Sea otter	V	36.5	5	M
Hughes et al. (2016)	<i>Z. marina</i> , Eelgrass	P	<i>E. lutris</i> , Sea otter	V	36.5	5	Pr
Lewis and Anderson (2012)	<i>Z. marina</i> , Eelgrass	P	<i>Micrometrus minimus</i> , Dwarf perch; <i>Paralabrax clathratus</i> , Kelp bass	V	32.4	4	Pr
Lu et al. (2018)	<i>Suaeda heteroptera</i> , Seepweed	P	Seabirds	V	40.5	3	M
McClanahan and Muthiga (2016)	Calcifying algae	Al	<i>Balistapus undulatus</i> , Orange-lined triggerfish	V	−9.3	4	M
Moreno et al. (1986)	<i>Chthamalus scabrorsus</i> , Barnacle; <i>J. cirratus</i> , Barnacle	An	<i>C. concholepas</i> , Rock snail	I	−39.2	3	M
Mumby et al. (2006)	Coral	An	<i>Epinephelus striatus</i> , Nassau grouper	V	24.2	3	M
Sharp and Angelini (2020)	<i>Spartina alterniflora</i> , Smooth cordgrass	P	Birds	V	30	3	Pr
Shears and Babcock (2003)	<i>Carpophyllum maschalocarpum</i> , Common flapjack; <i>Ecklonia radiata</i> , Spiny kelp	Al	<i>Pagrus auratus</i> , Australasian snapper	V	−36.2	3	M

Note: Foundational species (FS) types: AN, animal; AL, algae; P, plant. Top predator (TP) types: I, invertebrate; V, vertebrate. Manipulation (Manip): M, mensurative; Pr, pressed.

species when top predators were present in comparison with the control, no/low top predator treatment.

Predator effects

In this analysis of experimental restoration efforts in coastal systems, the addition of top predators generally increased the success of foundation species recovery. This indirect effect of predators (or TC) occurred because predators decreased consumer abundances (direct effect; negative LRR) which in turn increased foundation species abundances (indirect effect; positive LRR; Figure 1; $p < 0.001$). Contrary to our prediction, the magnitude of the predator effects on the consumers (direct) and foundation species (indirect) was not significantly different (Figure 1; $p = 0.2837$). While our initial analysis suggests that the indirect effect of top predators consistently facilitates the restoration of foundation species, the magnitude of the response is highly variable, as seen in the range of

LRR in Figure 1, and could be explained by both ecological (species identity, predator magnitude, and predator addition technique) and methodological (location and duration of experiment) factors.

Context dependence

Ecological factors

Variability in the effect of predators on restoration success was significantly influenced by taxonomic identity of the top predator, but not of the foundation species (Figure 2). Regardless of their identity (algae, animal, or true plant), foundation species increased by ~10 folds in experiments that simultaneously increased top predators versus those that did not (indirect effect; Figure 2a; power = 33%; $p = 0.515$). Inversely, the identity of the top predator was found to have significant effects on restoration outcomes: foundation species recovery was 3.5× greater when the

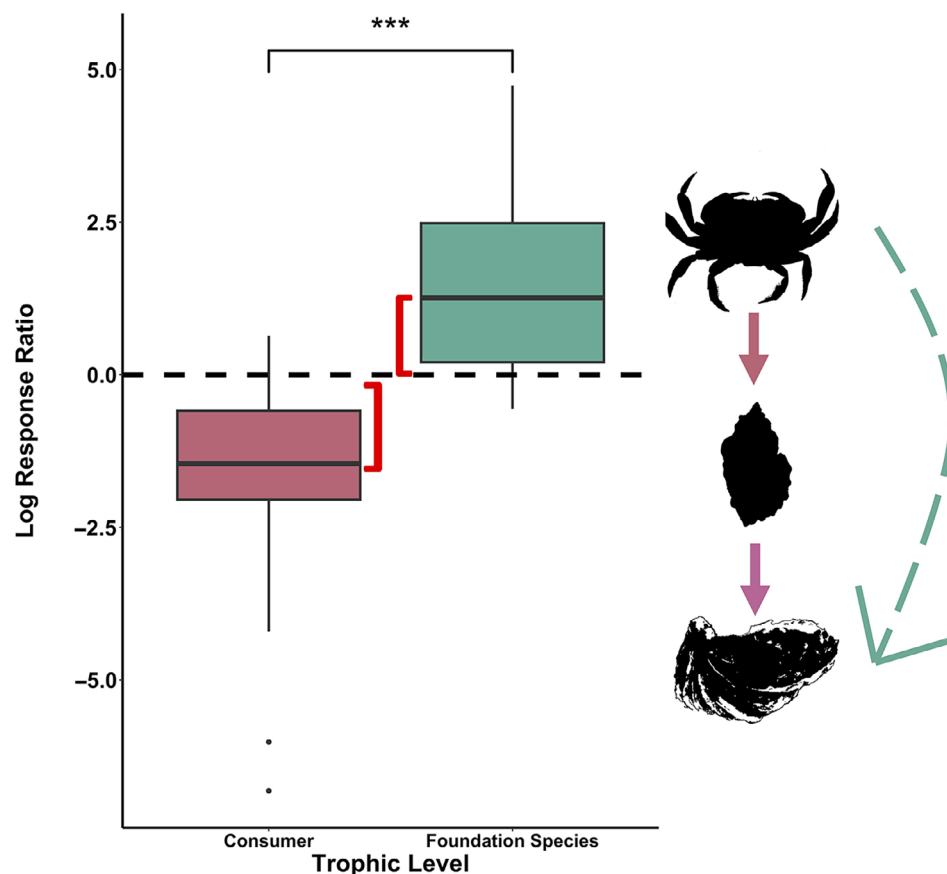


FIGURE 1 The effects of top predator addition relative to control on the log response ratio (LRR) of consumers (LRR = -1.81 ; $n = 18$) and foundation species (LRR = 1.57 ; $n = 18$) with box plots showing median values, box edges showing quartiles and error bars showing SE, and dots indicating outlier values. Pink color indicates direct effect of top predator addition and green indicates indirect effect. Asterisks denote a significant difference between the responses at p -value < 0.001 . The red vertical brackets represent mean magnitude of responses, which were not significantly different between consumers ($|LRR| = 1.81$; $n = 18$) and foundation species ($|LRR| = 1.81$; $n = 18$; $p = 0.340$). Line drawings from top to bottom by H. Eyster, M. Dahriel, T. Madea (Creative Commons, 2022).

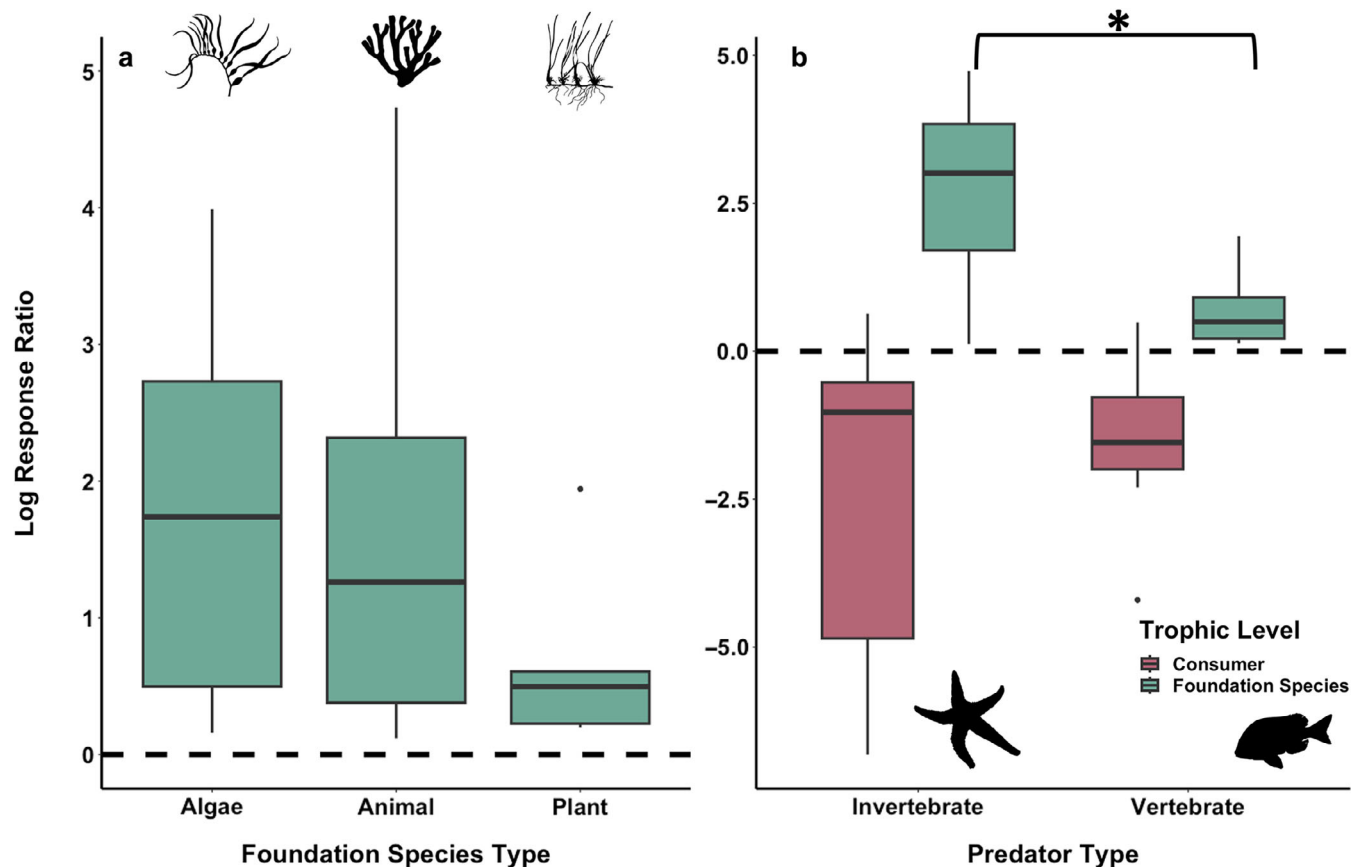


FIGURE 2 (a) Effect of top predator addition relative to control on log response ratio (LRR) of algae (LRR = 1.76; $n = 7$), animal (LRR = 1.69; $n = 6$), and true plant (LRR = 0.69; $n = 5$) foundation species with box plots showing median values, box edges showing quartiles, error bars showing SE, and dots indicating outlier values ($p = 0.5152$). Green color indicates indirect effect of top predator. Line drawings from left to right by H. Eyster, M. Kodis. (b) The effect of invertebrate and vertebrate predators on consumer (I LRR = -2.45; $n = 6$; V LRR = -1.45; $n = 11$; $p = 0.96$) and foundation species LRR (I LRR = 2.71; $n = 6$; V LRR = 0.69; $n = 11$), asterisk denotes significance at the $p < 0.05$ level. Green color indicates direct effect of top predator addition and pink indicates indirect effect. Line drawings from left to right by H. Hillewaert and T. M. Keesey, E. Schumacher (Creative Commons, 2022).

manipulated predators were invertebrate as opposed to vertebrate predators (indirect effect; Figure 2b; $p = 0.048$). However, invertebrate and vertebrate predators exerted similar control on consumer abundance (direct effect; Figure 2b; $p = 0.9618$). The invertebrate predators included cancrid crabs, carnivorous snails, rock lobsters, and seastars, while birds, fish, and sea otters dominated the vertebrate predator category.

Although the identity of the top predator influenced restoration success, the magnitude of predator change from treatment to controls did not elicit a significant effect on foundation species recovery (indirect effect; McFadden's $R^2 < 0.001$; $p = 0.415$). The methodology of these predator addition restoration experiments varied in two distinct ways. The first experimental approach consisted of 12 studies that used a mensurative, passive manipulation of top predator abundance due to cessation of harvesting predators or government intervention (11 of which were designated MPAs and one was a Ramsar

designated saltmarsh). The second experimental approach concerned six studies that used a pressed, explicit addition of top predators, approach. When top predators were allowed to recover naturally (mensurative), foundation species recovered 3.6 times greater than they did in the pressed studies, although this trend is not significant due to low statistical power, 59% (indirect effect; Figure 3; $p = 0.083$). In order to achieve significance at the 0.05 level, 20 studies per category would be required given the variance in these studies.

Methodological factors

In addition to ecological factors, methodological drivers, such as duration and location of the experiment, had variable effects on top predator addition as a foundation species restoration technique. In the Northern Hemisphere, the effect of latitude on predator-mediated recovery

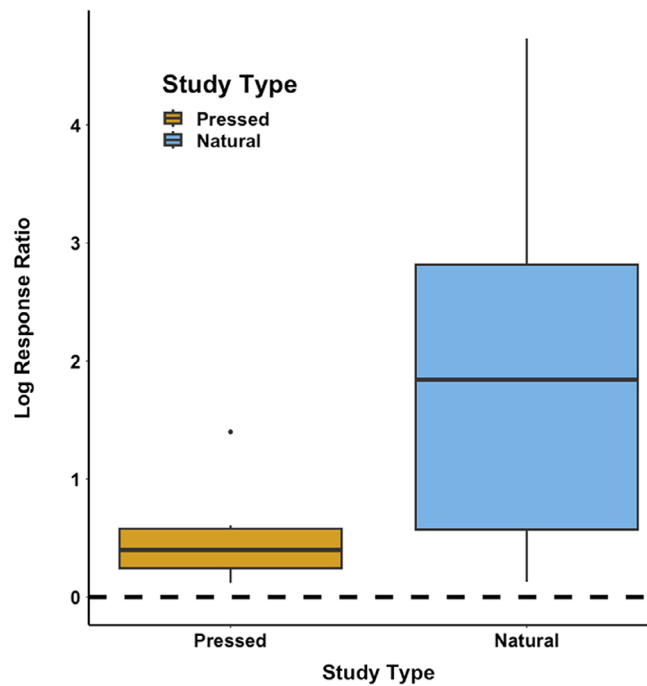


FIGURE 3 Foundation species log response ratio (LRR) in response to predator addition relative to control to pressed (yellow; LRR = 0.52; $n = 6$) and mensurative (blue; LRR = 1.90; $n = 12$) study designs with box plots showing median values, box edges showing quartiles, error bars showing SE, and dots indicating outlier values ($p = 0.08$).

(vs. restoration without manipulated predators) of foundation species was dependent on whether the manipulation was pressed (24.2–40.5° N) or mensurative (24.9–47.2° N; $p = 0.031$). This significant interaction was mainly driven by latitude having almost no effect on foundation species recovery in mensurative restoration experiments (Figure 4a; $p = 0.03$; McFadden's $R^2 = 0.07$; $p = 0.670$). However, in pressed experiments, increasing distance from the equator attenuated the indirect benefit of top predator addition on foundation species recovery (McFadden's $R^2 = 0.557$; $p = 0.155$). In the Southern Hemisphere, one outlier had to be removed at 9.3° S from the following analysis. No pressed experiments occurred in the Southern Hemisphere (33.3–42.4° S) and, similarly to the Northern Hemisphere, latitude had little to no effect on foundation species recovery in mensurative experiments (Figure 4b; McFadden's $R^2 = 0.07$; $p = 0.535$). Between the two manipulation types, duration length and the impact of study duration were significantly different, the maximum duration of the pressed studies was 120 days, while the mensurative studies lasted as long as 27.4 years ($p = 0.001$; $p = 0.011$). However, both pressed and mensurative experiments showed the same trend: increased duration of experiments suppressed foundation species response to top

predator addition (Figure 4c,d; McFadden's $R^2 = 0.214$; $p = 0.098$; McFadden's $R^2 = 0.807$; $p = 0.087$).

DISCUSSION

In restoration experiments, foundation species recovered more in projects that simultaneously enhanced the abundance of top predators. While these indirect benefits were due to the direct effect of top predators reducing the abundance of organisms that consume the foundation species, the strength of this indirect effect was highly variable. Thus, not only is there a larger role for predators to play in foundation species restoration, but there are several tractable attributes, such as predator identity, manipulation type, and time, that can be harnessed by practitioners to increase the magnitude of the indirect predator effect and, therefore, foundation species recovery.

Direct and indirect effects

The addition of top predators to enhance the success of foundation species restoration is an under-explored topic, and as such, there were relatively few studies to include in this meta-analysis. Accordingly, there was high variability in the strength of both the direct and indirect effects of predator inclusion within our results of foundation species recovery, despite classic ecological theory suggesting that direct effects should be stronger and less variable than indirect effects (Abrams, 1995; Wootton, 1994). While there was equal strength and variation in the response of both consumers (direct effect) and foundation species (indirect effect) to top predator addition, the motivation for our analysis, and thus that of the included publications, was on the indirect effects on foundation species recovery (Figure 1). As such, there was less emphasis within the included publications on the direct effects of top predators on consumers, such as whether predator effects were consumptive or nonconsumptive. The difference between these two pathways could influence the foundation species differently. As seen in Schmitz and Suttle (2001), density-mediated and trait-mediated indirect effects have different consequences for the basal resource despite the same top predator and intermediate consumer. This unexplained variability within the direct effects could be driving the variability seen in the indirect effects. Ultimately, we are interested in the foundation species recovery, but understanding the context-dependence of the consumer response could help to reduce variability in restoration outcomes.

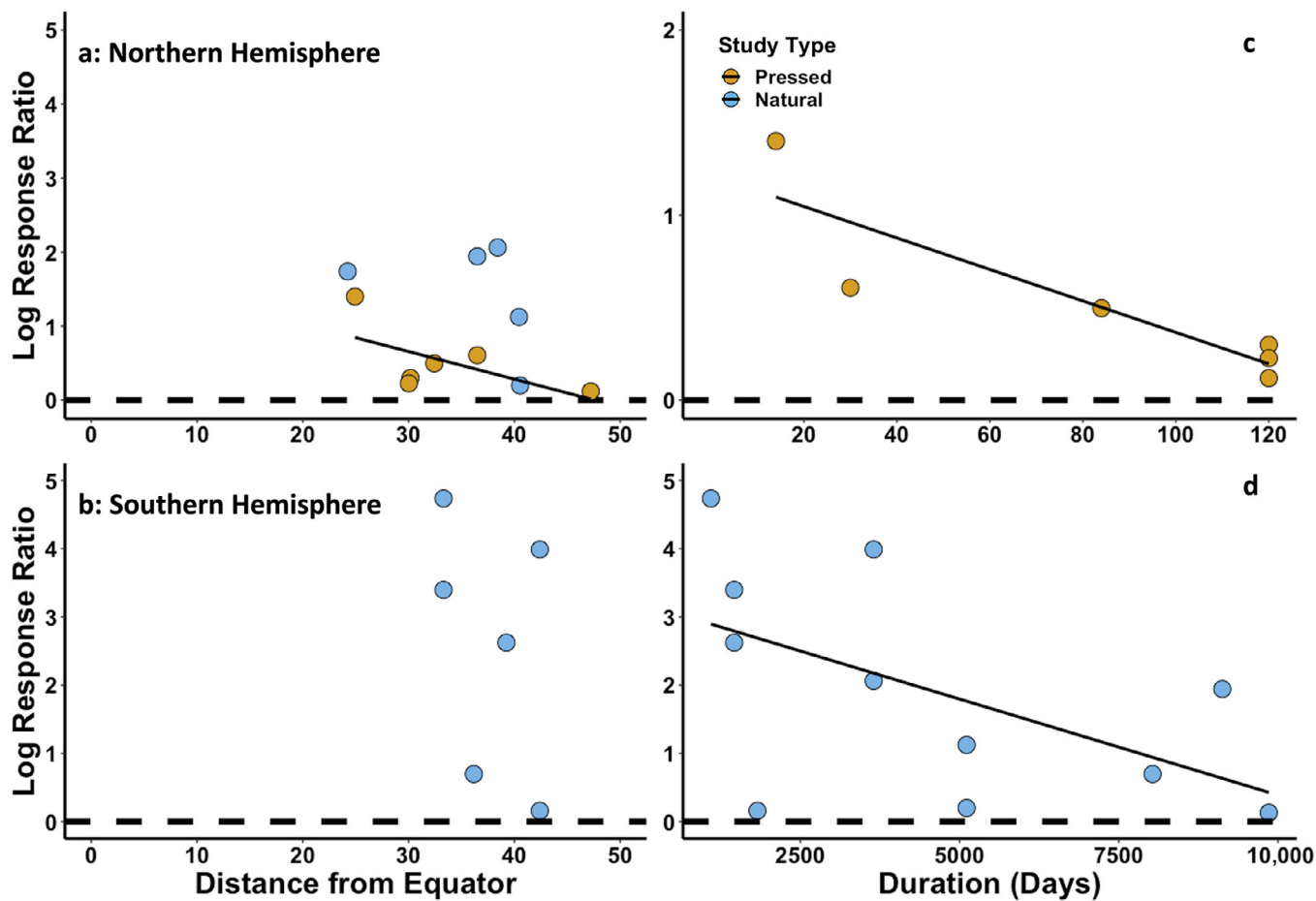


FIGURE 4 (a) The relationship between latitude and foundation species log response ratio (LRR) relative to control in the Northern Hemisphere in pressed (yellow) and mensurative (blue) studies (McFadden's $R^2 = 0.557$; $p = 0.155$; McFadden's $R^2 = 0.07$; $p = 0.670$). Generalized linear model regression line indicates trend. (b) The effect of latitude on foundation species LRR in the Southern Hemisphere in mensurative studies (blue; McFadden's $R^2 = 0.07$; $p = 0.389$). (c, d) The effect of study duration, in days, on foundation species LRR in pressed (yellow) and mensurative (blue) study designs (McFadden's $R^2 = 0.807$; $p = 0.087$; McFadden's $R^2 = 0.215$; $p = 0.098$). The generalized linear model regression line indicates trend.

Interestingly, invertebrate predators elicited a greater response in foundation species recovery than did vertebrate predators (Figure 2). In a recent review of the context-dependence of nonconsumptive predator effects, the authors highlighted that predators with large habitat domains are less likely to elicit both direct and indirect effects in cascading trophic levels (Schmitz, 2008; Wirsing et al., 2021). As vertebrate predators are generally larger, this discrepancy in foundation species response could be due to the fact that vertebrate predators, such as sea otters and birds, are more difficult to manipulate, are less numerous, have larger habitat domains, and require a more extensive permitting process than invertebrate predators such as sea stars and crabs. However, within the pressed experiments included in this analysis, the main vertebrate predators manipulated were small fish, which are similar in size to invertebrates and thus feasible to contain via caging. While

small fish and invertebrate predators are similar in size, they may respond to the effects of caging differently. Small fish may hunt over a larger area and have a higher metabolic rate than invertebrates and, therefore, become more stressed within experimental cages than the invertebrates, which could potentially dampen their predation effects (Englund, 1997; Gillooly et al., 2001). Future work should aim to disentangle whether our result is due to a fundamental difference in vertebrate and invertebrate predation or if the discrepancy is due to study artifacts.

Previous work has predicted that smaller, more controlled experiments result in stronger responses than larger, natural experiments (Englund & Cooper, 2003; Sagarin & Pauchard, 2009). A recent review of top predator effects on marine algae and plants found no difference in effect size between pressed and mensurative studies (Eger & Baum, 2020). Contrary to predictions, the foundation species response in our analysis was 3.5×

greater in mensurative publications than pressed studies. The increased foundation species response in mensurative approaches is a noteworthy outcome because few MPAs are created to specifically increase foundation species; they are typically designated to protect predatory species (Woodcock et al., 2017). Our results highlight the idea that MPAs have great potential for concurrent restoration of both predatory and foundation species. Results from recent modeling studies show promising results for kelp restoration within previously designated MPAs (Hopf et al., 2022). We suggest that that future implementation of MPAs should incorporate a larger ecosystem-wide approach and explicitly consider the benefits for foundation species as well as the target species, which are typically of fisheries interest.

According to our results, any such future research on foundation species restoration in MPAs will also need to consider the timing of MPA designation. A growing body of work suggests the effects of MPA designation decline over time (Magdaong et al., 2014). In particular, we found that foundation species LRR in mensurative studies declined as study duration increased. It has been demonstrated that due to algae's rapid growth rate, it rebounds quickly to reduced consumer pressure (Ramus, 1992; Reed et al., 2009). In our meta-analysis, we only used the last data point available in any study. So, while we cannot definitively say within a single mensurative experiment that the foundation species declined over time, when we analyzed study duration, the longer experiments tended to have lower foundation species LRR; however, all LRRs were still positive. Even though few pressed restoration experiments explicitly incorporated top predator addition (six publications), the influence of duration was also important for pressed experiments. Nevertheless, the very modest number of pressed experiments with predator additions also highlights a significant knowledge gap and the need for many more experimental studies involving the inclusion of top predators when experimentally examining the restoration of foundation species.

Although our findings illuminate that top predator addition is a broadly successful technique and should be explicitly incorporated into future foundation species restoration efforts, the latitude of the restoration effort will influence the effectiveness of harnessing top-down control for the purposes of restoration. As observed in general ecological studies for the strength of predation (Coley & Aide, 1991; Pennings et al., 2009), the strength of predator-mediated recovery of foundation species declines with increasing latitude, which could be due to increased abiotic stress and decreased species richness. Even though our results showed no statistically significant effect of latitude, they did show a general trend of

decreasing effect of predator control with increasing latitude in pressed experiments and no detectable effect in mensurative studies. The lack of statistical significance might be due to gaps in the data; while studies occurred in both hemispheres, there is only one study included in our meta-analysis near the equator, 9.3° S, and it had to be removed from the latitude analysis due to being a statistical outlier. In addition, no pressed studies were recorded in the Southern Hemisphere. While restoration experiments at higher latitudes tended to have reduced response to top predator addition, the LRR were still positive at all latitudes; thus, it is still a viable restoration technique at any latitude, but may need to be augmented with additional strategies at certain locations.

As the foundation species included in this meta-analysis were broad (algae, animals, and plants), we predicted that foundation species identity may be a source of variability in foundation species response. Regardless of foundation species identity, we found that each type responded positively to top predator addition. Our results are in accordance with TC theory in that TCs are found in a diverse array of ecosystems, regardless of the identity of basal trophic level species (Pace et al., 1999). We did not find support for the generation time theory that shorter generation time organisms (plants or algae) are more conducive to TCs than longer generation time organisms, such as coral or oysters. A recent review of the effects of top predator declines on marine algae and plants found kelp and salt marsh plants responded most strongly and seagrasses had the weakest response to predator losses (Atwood & Hammill, 2018). Due to sample size limitations, we could not examine differences among plant types. This result suggests that subtypes within our broad categories may respond differently to top predator addition as a restoration technique and merits further investigation.

In addition to predator identity, we examined whether predator density influenced foundation species recovery. Predator presence caused a significant increase in foundation species recovery; however, there was no additive effect of increased predator density. As mentioned previously, there was little emphasis within included publications regarding whether top predator effects were consumptive or nonconsumptive. This result loosely indicates that the effects may generally be more nonconsumptive than consumptive as any predator presence resulted in an increase in foundation species.

Conclusion

Top predator addition remains a nascent technique in the foundation species restoration field, shown by the modest

number of studies in our analysis. Despite the small sample size, we found very strong effects of top predator addition that resulted in significant enhancement of foundation species recovery in a broad variety of contexts. While there are some significant predictors of success, including aspects of the study design and the top predator identity, top predator addition is a generally very successful restoration technique for foundation species. The need for foundation species restoration is imperative to promote biodiversity and provide ecosystem services worldwide. Our results support the idea that top predators should be explicitly included in foundation species restoration projects and that MPAs should explicitly include beneficial effects on foundation species in the development of MPA management policies.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data and code (Heineke, 2023) used are available at the Northeastern University Digital Repository Service: <http://hdl.handle.net/2047/D20476524>.

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